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🎯 Quiz: Model Fine-Tuning & LLM Optimization

🧠 Section 1: Conceptual MCQs

1. What is model fine-tuning?
A. Training a model from scratch
B. Adjusting a pre-trained model on new data
C. Only changing prompts
D. Increasing model size

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2. Which scenario is best suited for prompting instead of fine-tuning?
A. Domain-specific chatbot
B. Repetitive structured output task
C. One-time query with general knowledge
D. Training on private dataset

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3. What is the main advantage of fine-tuning over prompting?
A. Faster inference always
B. Better control over task-specific outputs
C. No need for data
D. Lower memory usage

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4. What does LoRA stand for?
A. Low Rank Adaptation
B. Linear Representation Adjustment
C. Layer Optimization and Reuse Algorithm
D. Learning Rate Adjustment

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5. In LoRA, what is actually trained?
A. Entire model weights
B. Only input tokens
C. Small low-rank matrices
D. Output layer only

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6. Why is LoRA considered parameter-efficient?
A. It reduces dataset size
B. It updates only a small subset of parameters
C. It removes layers
D. It compresses input

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7. What is the role of a tokenizer in fine-tuning?
A. Train model weights
B. Convert text into tokens
C. Evaluate model
D. Store embeddings

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8. Which step comes first in a fine-tuning workflow?
A. Model evaluation
B. Data collection & cleaning
C. Training
D. Deployment

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9. What is overfitting in fine-tuning?
A. Model performs well on all data
B. Model memorizes training data but fails on new data
C. Model ignores training data
D. Model becomes faster

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10. Which metric is commonly used for classification tasks?
A. BLEU score
B. Accuracy

C. Perplexity
D. Latency

🔍 Section 2: True / False

11. Fine-tuning always requires large datasets. -> FALSE
12. Prompting does not change model weights. -> TRUE
13. LoRA modifies all layers of the model. -> FALSE
14. Validation data is used to tune hyperparameters. ->TRUE
15. Fine-tuned models can outperform base models on niche tasks. ->TRUE

🧩 Section 3: Scenario-Based Questions

16. Scenario:
You need a chatbot specialized in legal documents with consistent tone and formatting.
👉 What should you choose?
A. Prompting
B. Fine-tuning
C. Zero-shot learning
D. Random initialization

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17. Scenario:
You want quick results for generating marketing slogans without training.
👉 Best approach?
A. Fine-tuning
B. Prompt engineering
C. LoRA
D. Training from scratch

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18. Scenario:
You have limited GPU resources but want to adapt a large model.
👉 Best method?
A. Full fine-tuning

B. LoRA
C. Increase batch size
D. Add more layers

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19. Scenario:
Your model performs well on training data but poorly on test data.
👉 What is the issue?
A. Underfitting
B. Overfitting
C. Data leakage
D. Tokenization error

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20. Scenario:
You are comparing two models for summarization.
👉 Which metric is most relevant?
A. Accuracy
B. BLEU / ROUGE
C. Latency only
D. Token count

⚙️ Section 4: Short Answer

21. Explain one difference between prompting and fine-tuning.

Prompting uses instructions without changing model weights, while fine-tuning updates model weights using training data.

22. What is one benefit of using LoRA?

It reduces the number of parameters that need to be trained.

23. List any two steps in a fine-tuning workflow.

Data collection & cleaning, and training.

24. What is the purpose of validation data?

To tune hyperparameters and check model performance on unseen data.

25. Name one metric used to evaluate LLMs.

Accuracy (or BLEU/ROUGE).

🧪 Section 5: Practical Thinking

26. You are fine-tuning a model and loss is not decreasing. What could be one reason?

Learning rate may be too high or too low (or poor-quality data).

27. Why might fine-tuning be expensive?

Because it requires high compute (GPUs), time, and large datasets

28. When should you avoid fine-tuning?

When the task can be handled well with prompting or is simple/one-time.

29. What happens if tokenizer is mismatched with the model?

The model may produce incorrect, inconsistent, or meaningless outputs.

30. How does LoRA reduce computational cost?

By training only small low-rank matrices instead of the full model weights.